1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

AB
C

D

E

FGH

IJK
LM
N

O

PQ
R

S

TU

VW
X

YZ
[

\

]

^_

`a
b

cd
e

fg
h

i

jk

lm
n

o

AB
C

D

E

FGH

IJK
LM
N

O

PQ
R

S

TU

VW
X

YZ
[

\

]

^_

`a
b

cd
e

fg
h

i

jk

lm
n

o Date: 2026-08-20

KiCad E.D.A. 10.0.4

Size: User Rev: dev
Id: 1/1

Title: usb_controlled_debug_hub_2a

File: usb_controlled_debug_hub_2a.kicad_sch

Sheet: /

R_NONREM0

10kΩ

GND_A1 A1

VBUS_A4 A4 CC1 A5 DP_A6 A6 DM_A7 A7

SBU1 A8

VBUS_A9 A9 GND_A12 A12 GND_B1 B1 VBUS_B4 B4 CC2 B5 DP_B6 B6 DM_B7 B7

SBU2 B8

VBUS_B9 B9 GND_B12 B12

SHIELD SH

J_DATA C165948 C_HUB_18

1uF

R_CC1

5.1kΩ

C_HUB_18PLL 1uF C_HUB_A4

100nF C_HUB_A3

100nF C_HUB_PLL 100nF C_HUB_A2

100nF R_HUB_RESET 10kΩ

C_HUB_CR_BULK

1uF C_HUB_A_BULK
1uF R_CFG0

10kΩ

C_HUB_CR_HF 100nF R_XTAL 1MΩ

C_XTAL1

18pF C_XTAL2

18pF

X1 1 GND1 2

X2 3 GND2 4 Y_HUB
C70590

C_PWR2_IN

100nF

C_DATA1

100nF

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 U_DATA1 C11355

G

1
S

2 D

3 Q_DATA1 C85047 R_DATA_OE1

10kΩ

C_DATA2

100nF

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 U_PWR2 C3662799

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 U_DATA2 C11355

R_SWAP1

100kΩ

R_CFG1

10kΩ

R_CFG2

10kΩ

R_SWAP4

100kΩ

R_NONREM1

10kΩ

R_SWAP3

100kΩ

R_SWAP5

100kΩ

R_SWAP7

100kΩ

R_SWAP6

100kΩ

R_SWAP2

100kΩ

R_DATA_OK1

10kΩ

R_AGG_A_ILIM

300Ω

C_BANK_A_HF 100nF C_AGG_A_DVDT 3.3nF C_BANK_B_BULK

22uF C_AGG_A_TIMER

6.8nF C_BANK_B_HF 100nF C_BANK_B_POLY 180uF

C_BANK_A_POLY 180uF C_BANK_A_BULK

22uF

R_AGG_B_ILIM

300Ω

C_PD_IN1

10uF C_PD_IN2

10uF

C_AGG_B_DVDT 3.3nF C_AGG_B_TIMER

6.8nF

IN1

1
IN2

2
IN3

3 GND1
4 GND2

5
EN_UVLO

6
ITIMER

7
ILIM

8
IMON

9 RETRY_DLY 10 NRETRY 11
LDSTRT 12 PG

13 GND3

14 DVDT 15

IN4

16 OUT1

17 OUT2

18 OUT3

19 OUT4

20 OUT5

21 OUT6

22 OUT7

23 OUT8

24

IN_POWERPAD

25 GND_POWERPAD

26 U_AGG_A C2155765 DN1_DM

1 DN1_DP

2 DN2_DM

3 DN2_DP

4

VDDA33_1 5 DN3_DM 6 DN3_DP 7 DN4_DM 8 DN4_DP 9
VDDA33_2 10 DN5_DM 11 DN5_DP 12 CFG_SEL2 13
LED_B7 14 PRT_SWP7 15
LED_B6 16 PRT_SWP6 17
LED_B5 18

TEST 19 PRTPWR4 20 OCS4_N 21 OCS3_N 22 PRTPWR3 23

VDD33CR 24

VDD18 25 PRTPWR2 26 OCS2_N 27 OCS1_N 28 PRTPWR1 29 PRTPWR5 30 PRT_SWP5 31 LED_B4 32 PRT_SWP4 33 GANG_EN 34 OCS5_N 35 PRTPWR7 36 OCS7_N 37 OCS6_N 38 PRTPWR6 39 NON_REM1 40 CFG_SEL0 41 CFG_SEL1 42 RESET_N 43

VBUS_DET 44 NON_REM0 45

VDD33 46 PRT_SWP3 47 BOOST1 48 PRT_SWP2 49 BOOST0 50 PRT_SWP1 51

VDDA33_3 52 DN6_DM 53 DN6_DP 54 DN7_DM 55 DN7_DP 56

VDDA33_4 57 UP_DM 58 UP_DP 59

XTAL2 60

XTAL1 61 VDD18PLL 62 RBIAS 63 VDD33PLL 64 EP_VSS 65 U_HUB

C478081

R_BUCK_B_EN_TOP

200kΩ

R_RBIAS

12kΩ

R_BUCK_B_EN_BOT 27.4kΩ

R_VBUS_BOT 100kΩ

R_VBUS_TOP

47kΩ

IO1 1

IO2 2 GND 3 U_ESD_UP

C3709087 R_CC2

5.1kΩ

C_HUB_A1

100nF C_HUB_DD
100nF

C_AGG_B_IN

100nF

IN1

1
IN2

2
IN3

3 GND1
4 GND2

5
EN_UVLO

6
ITIMER

7
ILIM

8
IMON

9 RETRY_DLY 10 NRETRY 11
LDSTRT 12 PG

13 GND3

14 DVDT 15

IN4

16 OUT1

17 OUT2

18 OUT3

19 OUT4

20 OUT5

21 OUT6

22 OUT7

23 OUT8

24

IN_POWERPAD

25 GND_POWERPAD

26 U_AGG_B

C2155765

C_AGG_A_IN

100nF

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4
SHIELD 5 J_PORT1 USB1130-15-A

C_BUCK_B_IN1

10uF C_BUCK_B_IN2

10uF

R_PWR_CMD2

10kΩ

R_DATA_CMD4

10kΩ

R_PWR_CMD4

10kΩ

R_I2C_SDA 4.7kΩ
R_PWR_CMD3

10kΩ

R_I2C_SCL 4.7kΩ
R_PWR_CMD1

10kΩ

R_DATA_CMD3

10kΩ

R_DATA_CMD2

10kΩ

R_DATA_CMD1

10kΩ

C_PWR_CTRL_IN

100nF

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 U_DATA3 C11355 C_DATA3

100nF C_PORT3_BULK

22uF

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4
SHIELD 5 J_PORT3 USB1130-15-A

R_ILIM_CTRL 187kΩ

IO1 1

IO2 2 GND 3 U_ESD3 C3709087 C_PORT3_HF 100nF

C_PWR_CTRL_OUT 1uF

1A 1
1B 2
1Y 3
2A 4
2B 5

2Y 6 GND 7

3Y 8

3A 9

3B 10

4Y 11

4A 12

4B 13

VCC 14 U_AND_PWR

C6053

R_DIS6P

10kΩ

GND 1
IN1 2
IN2 3
EN 4

ILIM 5 OUT1 6 OUT2 7

FAULT_N 8

EP 9 U_PWR_CTRL C130056

R_GANG

10kΩ

GPB0 1 GPB1 2 GPB2 3 GPB3 4 GPB4 5 GPB5 6 GPB6 7 GPB7 8

VDD 9

VSS 10 NC1 11
SCL 12

SDA 13 NC2 14 A0 15

A1 16

A2 17 RESET_N 18

INTB 19

INTA 20 GPA0 21 GPA1 22 GPA2 23 GPA3 24 GPA4 25 GPA5 26 GPA6 27 GPA7 28 U_EXP
C558584 R_DIS6N

10kΩ

R_DIS7P

10kΩ

R_DIS7N

10kΩ

VDD 1 GP0 2 GP1 3 RST 4 URX 5 UTX 6 GP2 7 GP3 8

SDA 9

SCL 10

VUSB 11 D_MINUS 12 D_PLUS 13

VSS 14 U_CTRL C130462

R_BOOST0

10kΩ

R_BOOST1

10kΩ

R_EXP_RESET 10kΩ

C_PWR_CTRL_OUT_HF 100nF C_CTRL_VDD
100nF R_CTRL_RESET 10kΩ

C_EXP_VDD

100nF C_CTRL_VUSB

330nF

1A 1
1B 2
1Y 3
2A 4
2B 5

2Y 6 GND 7

3Y 8

3A 9

3B 10

4Y 11

4A 12

4B 13

VCC 14 U_AND_DATA C6053

C_AND_DATA 100nF C_AND_PWR

100nF

R_DATA_OK2

10kΩ

R_ILIM1

1.4kΩ

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 U_PWR1 C3662799

C_PORT2_BULK

22uF C_PORT2_HF 100nF

IO1 1

IO2 2 GND 3 U_ESD2 C3709087

C_PWR3_IN

100nF R_PWR_EN3

10kΩ

R_DATA_OE4

10kΩ

R_PWR_EN1

10kΩ

R_DATA_OE2

10kΩ

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4
SHIELD 5 J_PORT2 USB1130-15-A

C_PORT1_HF 100nF

R_ILIM3

1.4kΩ

IO1 1

IO2 2 GND 3 U_ESD1 C3709087

G

1
S

2 D

3 Q_DATA2 C85047

C_PORT1_BULK

22uF C_PWR1_IN

100nF

G

1
S

2 D

3 Q_DATA3 C85047 R_DATA_OE3

10kΩ

R_PWR_EN2

10kΩ

R_ILIM2

1.4kΩ

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4
SHIELD 5 J_PORT4 USB1130-15-A

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 U_PWR4 C3662799 C_PORT4_HF 100nF

IO1 1

IO2 2 GND 3 U_ESD4 C3709087 C_PORT4_BULK

22uF

R_DATA_OK3

10kΩ

R_DATA_OK4

10kΩ

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 U_PWR3 C3662799

G

1
S

2 D

3 Q_DATA4 C85047 C_PWR4_IN

100nF R_ILIM4
1.4kΩ

C_HUB_RESET 1uF

C_DATA4

100nF

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 U_DATA4 C11355 R_PWR_EN4

10kΩ

C_PD_IN_HF 1uF

R_PD_UV_TOP_LO

22kΩ

IN1 1

IN2 2 NC1 3 NC2 4 NC3 5 P_IN 6 UVLO 7 OVP 8 GND 9 DVDT 10

ILIM 11 MODE 12

SHDN 13

IMON 14

FLT 15 PGOOD 16 NC4 17 OUT1 18 OUT2 19 NC5 20

EP 21 U_PD_IN

C1849461 R_PD_OV_BOT 51kΩ

R_PD_UV_MID

20kΩ

L_MAIN

C15269

R_FB_A_BOT 10kΩ

C_BUCK_A_OUT1

22uF

R_PD_IN_ILIM

5.9kΩ

C_MAIN_IN

10uF

R_PD_UV_TOP_HI 910kΩ

C_BUCK_B_OUT2

22uF

C_MAIN_OUT1

22uF

GND_A1 A1

VBUS_A4 A4 CC1 A5 DP_A6 A6 DM_A7 A7

SBU1 A8

VBUS_A9 A9 GND_A12 A12 GND_B1 B1 VBUS_B4 B4 CC2 B5 DP_B6 B6 DM_B7 B7

SBU2 B8

VBUS_B9 B9 GND_B12 B12

SHIELD SH

J_POWER

C165948

C_BST_B

100nF

R_PD_VBUS

10kΩ

C_BUCK_B_OUT1

22uF

FB 1
EN 2
VIN 3 GND 4

SW 5 BST 6 U_MAIN

C5248536 C_BST 100nF

R_FB_A_TOP

75kΩ

C_PD_IN_DVDT 3.3nF

C_BUCK_A_IN1

10uF

R_PD_VDD

1kΩ

C_MAIN_OUT2

22uF

VDD 1 CFG2 2 CFG3 3 DP 4 DM 5 CC2 6 CC1 7

VBUS 8 CFG1 9 PG 10

EP_GND 11 U_PD

C970725

R_FB_B_TOP

75kΩ

F_PD

C14165

C_BST_A 100nF R_BUCK_A_EN_TOP

200kΩ

EN 1
FB 2

AGND 3 PG 4 NC 5

SW 6 BOOT 7

VIN 8 PGND 9 MODE 10 U_BUCK_B
C841386

C_BUCK_A_OUT2

22uF

L_BUCK_B

C17700166

GND1 1 GND2 2 GND3 3

IN1 4 IN2 5

IN3 6

EP_GND 7 D_PD_TVS C523793

R_FB_B_BOT 10kΩ

R_BUCK_A_EN_BOT 27.4kΩ

C_BUCK_A_IN2

10uF

EN 1
FB 2

AGND 3 PG 4 NC 5

SW 6 BOOT 7
VIN 8 PGND 9 MODE 10 U_BUCK_A C841386

L_BUCK_A C17700166

C_PD_VDD

1uF

P1_PORT_N

3V3_MAIN

PWR_EN1 VBUS1_SW
PWR_EN2

P1_PORT_P VBUS_PD

P2_HUB_P

P2_HUB_N

VBUS_PD

P5V_A_PROTECTED HUB_OCS3_N

HUB_OCS4_N VBUS2_SW

P5V_A_PROTECTED

VBUS1_SW

ILIM1 P5V_A_PROTECTED

FB_A

P2_PORT_N P2_PORT_N

DATA_OK2

HUB_OCS2_N

P5V_BANK_A

P2_PORT_P

DATA_OE2_N

ILIM2

P5V_B_PROTECTED

PWR_EN3

ILIM4

P2_PORT_P

PWR_EN1 P5V_A_PROTECTED

P5V_BANK_B

P5V_BANK_B

DATA_OK2 DATA_OE2_N

FB_B

P1_PORT_N

P1_PORT_P

ILIM1 VBUS1_SW

P5V_BANK_A VBUS2_SW

P2_PORT_N
P2_PORT_P

UP_HUB_P

VBUS_PD_PROTECTED

PD_CC1

3V3_MAIN VBUS_PD_RAW

3V3_MAIN VBUS_PD_RAW

XTAL2 VBUS_PD_RAW

PD_UV_TOP_MID HUB_RESET_N

UP_HUB_N

DATA_CC2 DATA_CC1 USB_UP_VBUS DATA_CC1 UP_HUB_P UP_HUB_N

PD_IN_OV

P3_HUB_N

USB_UP_VBUS UP_HUB_P UP_HUB_N MGMT_P

P1_HUB_N
P1_HUB_P

P2_HUB_N
P2_HUB_P

HUB_VDD18PLL

P3_HUB_P MGMT_N

3V3_MAIN

DATA_CC2

3V3_MAIN

3V3_MAIN

PD_CC2

HUB_SWAP7

P1_HUB_P

HUB_SWAP4 HUB_SWAP7 HUB_SWAP3 HUB_SWAP6

3V3_MAIN

P1_HUB_N

P1_PORT_P

P1_PORT_N

DATA_OK1

PD_IN_DVDT

DATA_OE1_N

DATA_OK1 DATA_OE1_N

PD_VDD

3V3_MAIN

3V3_MAIN

P5V_A_PROTECTED

P5V_A_PROTECTED VBUS_PD HUB_NONREM0

HUB_GANG

XTAL1 PD_IN_UV

3V3_MAIN

HUB_VDD18 HUB_CFG0HUB_SWAP1

ILIM2

HUB_CFG1 HUB_CFG2 HUB_NONREM1 3V3_MAIN

PD_IN_ILIM

HUB_SWAP5

3V3_MAIN HUB_SWAP2

FB_A

CTRL_RESET_N VBUS_CTRL

I2C_SDA

3V3_MAIN VBUS_CTRL EXP_RESET_N

EXP_RESET_N

PWR_CMD1 I2C_SDA VBUS_CTRL
HUB_PRTPWR4 PWR_CMD3 PWR_EN4 HUB_PRTPWR5 PWR_CMD4 PWR_EN2 DATA_OK1 DATA_CMD2 DATA_OK2

P5V_A_PROTECTED

PWR_CMD4

DATA_CMD4 PWR_EN1 DATA_CMD1 PWR_CMD3

BOOT_B SW_B

DATA_CMD3

PD_IN_DVDT

VBUS_PD_PROTECTED

P5V_A_PROTECTED

P5V_A_PROTECTED

ILIM_CTRL

ILIM_CTRL HUB_OCS1_N VBUS_CTRL VBUS_CTRL CTRL_VUSB_3V3

I2C_SCL DATA_CMD3 DATA_CMD4

I2C_SDA CTRL_VUSB_3V3 VBUS_CTRL

I2C_SCL PWR_CMD2 PWR_CMD3 HUB_DIS7_P

DATA_CMD2 MGMT_N MGMT_P

VBUS_CTRL VBUS_CTRL VBUS_PD_PROTECTED

3V3_MAIN

HUB_BOOST1 HUB_DIS7_N

BUCK_A_EN

PD_VBUS_SENSE

PWR_EN3 PWR_EN4 DATA_OK3 DATA_CMD3 DATA_OK4

DATA_CMD4

BOOT_A

HUB_DIS6_N

PD_IN_ILIM

DATA_CMD2

HUB_BOOST0 HUB_PRTPWR1

3V3_MAIN HUB_DIS6_P HUB_GANG

BUCK_A_EN

PWR_CMD4 DATA_CMD1

HUB_OCS5_N

DATA_OK4 VBUS4_SW

P4_PORT_P

P4_PORT_P

P4_HUB_N VBUS_PD_PROTECTED

P4_HUB_P

DATA_OE4_N

PWR_EN4

P4_PORT_N

P5V_B_PROTECTED

PWR_EN4

BST_3V3

DATA_OE3_N

DATA_OK3 P3_PORT_P

P3_PORT_P

P3_PORT_N VBUS3_SW

ILIM3

BOOT_A SW_A

PWR_EN3

3V3_MAIN

P5V_BANK_A

VBUS3_SW VBUS3_SW
ILIM3 P5V_B_PROTECTED

VBUS2_SW

DATA_OE4_N DATA_OK4

3V3_MAIN

P5V_B_PROTECTED

P5V_B_PROTECTED VBUS_PD_PROTECTED

SW_A

P4_PORT_N

PWR_CMD2 VBUS_PD_PROTECTED

P5V_BANK_B

P3_PORT_P

P3_PORT_N

CTRL_RESET_N

HUB_PRTPWR2 PWR_CMD1 PWR_EN1 HUB_PRTPWR3

DATA_OK3

3V3_MAIN

BOOT_B

PWR_EN2 PWR_EN3

3V3_MAIN

PWR_CMD1 PWR_CMD2 DATA_CMD1 VBUS4_SW

3V3_MAIN

3V3_MAIN

P3_HUB_N
P3_HUB_P

PWR_EN2

3V3_MAIN

ILIM4 P4_PORT_P

P4_PORT_N

P5V_B_PROTECTED

BUCK_A_EN

VBUS4_SW

SW_3V3 BST_3V3 SW_3V3
FB_A

P3_PORT_N

DATA_OE3_N

HUB_DIS6_P HUB_VDD18PLL

SW_B VBUS_PD_PROTECTED VBUS_PD

FB_B

BUCK_B_EN

PD_VDD VBUS_PD_PROTECTED HUB_DIS7_N

RBIAS HUB_DIS6_N

HUB_SWAP1 HUB_BOOST0

3V3_MAIN

HUB_DIS7_P UP_HUB_N

UP_HUB_P

P5V_BANK_B

HUB_RESET_N

HUB_SWAP6 HUB_OCS5_N

XTAL1 HUB_SWAP2 HUB_BOOST1 HUB_SWAP3 HUB_NONREM0 HUB_VBUS_SENSE

P5V_BANK_A

HUB_CFG1 HUB_CFG0 HUB_NONREM1 HUB_VBUS_SENSE BUCK_B_EN 3V3_MAIN

XTAL1
XTAL2

P5V_B_PROTECTED

PD_IN_UV

AGG_A_TIMER AGG_A_DVDT

P5V_A_PROTECTED

P5V_BANK_A

AGG_A_TIMER
AGG_A_ILIM

P5V_BANK_A

PD_IN_OV

P5V_B_PROTECTED

AGG_B_ILIM

P5V_BANK_B

AGG_B_DVDT VBUS_PD

AGG_B_TIMER AGG_B_DVDT

VBUS_PD_PROTECTED VBUS_PD_PROTECTED

AGG_B_TIMER

P5V_BANK_A P5V_A_PROTECTED

PD_CC1 VBUS_PD

PD_VBUS_SENSE

P5V_BANK_B VBUS_PD

AGG_B_ILIM

HUB_SWAP4

P5V_A_PROTECTED

PD_CC2 PD_PROTO

P5V_B_PROTECTED

P5V_A_PROTECTED

P5V_BANK_B

AGG_A_DVDT AGG_A_ILIM

USB_UP_VBUS P4_HUB_N

HUB_PRTPWR5 HUB_CFG2 HUB_OCS2_N

HUB_OCS1_N

HUB_VDD18 P4_HUB_P HUB_SWAP5 HUB_PRTPWR3 HUB_OCS3_N

HUB_OCS4_N

HUB_PRTPWR2 HUB_PRTPWR4 HUB_PRTPWR1 3V3_MAIN

U_BUCK_B

C841386

EN 1
FB 2

AGND 3 PG 4 NC 5

SW 6 BOOT 7

VIN 8 PGND 9 MODE 10

C_BUCK_A_IN1

10uF C_BUCK_A_IN2

10uF R_BUCK_A_EN_BOT 27.4kΩ

R_PD_IN_ILIM

5.9kΩ

C_MAIN_OUT1

22uF C_MAIN_OUT2

22uF

C_PD_IN_DVDT 3.3nF

C_BUCK_B_OUT1

22uF

U_MAIN

C5248536

FB 1
EN 2
VIN 3 GND 4

SW 5 BST 6 C_MAIN_IN

10uF U_BUCK_A C841386

EN 1
FB 2

AGND 3 PG 4 NC 5

SW 6 BOOT 7

VIN 8 PGND 9 MODE 10 R_FB_A_BOT 10kΩ
C_BUCK_B_OUT2

22uF

C_BUCK_A_OUT1

22uF C_BUCK_A_OUT2

22uF

R_FB_B_BOT 10kΩ

C_PD_VDD

1uF D_PD_TVS C523793

GND1 1 GND2 2 GND3 3

IN1 4 IN2 5

IN3 6

EP_GND 7 R_PD_OV_BOT 51kΩ

J_POWER

C165948

GND_A1 A1

VBUS_A4 A4 CC1 A5 DP_A6 A6 DM_A7 A7

SBU1 A8

VBUS_A9 A9 GND_A12 A12 GND_B1 B1 VBUS_B4 B4 CC2 B5 DP_B6 B6 DM_B7 B7

SBU2 B8

VBUS_B9 B9 GND_B12 B12

SHIELD SH C_PD_IN_HF 1uF U_PD

C970725

VDD 1 CFG2 2 CFG3 3 DP 4 DM 5 CC2 6 CC1 7

VBUS 8 CFG1 9 PG 10

EP_GND 11 U_PD_IN

C1849461

IN1 1

IN2 2 NC1 3 NC2 4 NC3 5 P_IN 6 UVLO 7 OVP 8 GND 9 DVDT 10

ILIM 11 MODE 12

SHDN 13

IMON 14

FLT 15 PGOOD 16 NC4 17 OUT1 18 OUT2 19 NC5 20

EP 21 C_PD_IN1

10uF C_PD_IN2

10uF

C_AGG_B_DVDT 3.3nF C_AGG_B_TIMER

6.8nF R_AGG_B_ILIM

300Ω

U_AGG_A C2155765

IN1

1
IN2

2
IN3

3 GND1
4 GND2

5
EN_UVLO

6
ITIMER

7
ILIM

8
IMON

9 RETRY_DLY 10 NRETRY 11
LDSTRT 12 PG

13 GND3

14 DVDT 15

IN4

16 OUT1

17 OUT2

18 OUT3

19 OUT4

20 OUT5

21 OUT6

22 OUT7

23 OUT8

24

IN_POWERPAD

25 GND_POWERPAD

26 C_BANK_A_HF 100nF C_AGG_A_DVDT 3.3nF C_BANK_B_BULK

22uF C_AGG_A_TIMER

6.8nF R_AGG_A_ILIM

300Ω

C_BANK_B_HF 100nF C_BANK_B_POLY 180uF

C_BANK_A_POLY 180uF C_BANK_A_BULK

22uF

U_AGG_B

C2155765

IN1

1
IN2

2
IN3

3 GND1
4 GND2

5
EN_UVLO

6
ITIMER

7
ILIM

8
IMON

9 RETRY_DLY 10 NRETRY 11
LDSTRT 12 PG

13 GND3

14 DVDT 15

IN4

16 OUT1

17 OUT2

18 OUT3

19 OUT4

20 OUT5

21 OUT6

22 OUT7

23 OUT8

24

IN_POWERPAD

25 GND_POWERPAD

26

C_AGG_A_IN

100nF

C_AGG_B_IN

100nF C_BUCK_B_IN1

10uF C_BUCK_B_IN2

10uFR_RBIAS

12kΩ

R_BUCK_B_EN_BOT 27.4kΩ

R_VBUS_BOT 100kΩ

U_HUB

C478081 DN1_DM

1 DN1_DP

2 DN2_DM

3 DN2_DP

4

VDDA33_1 5 DN3_DM 6 DN3_DP 7 DN4_DM 8 DN4_DP 9
VDDA33_2 10 DN5_DM 11 DN5_DP 12 CFG_SEL2 13
LED_B7 14 PRT_SWP7 15
LED_B6 16 PRT_SWP6 17
LED_B5 18

TEST 19 PRTPWR4 20 OCS4_N 21 OCS3_N 22 PRTPWR3 23

VDD33CR 24

VDD18 25 PRTPWR2 26 OCS2_N 27 OCS1_N 28 PRTPWR1 29 PRTPWR5 30 PRT_SWP5 31 LED_B4 32 PRT_SWP4 33 GANG_EN 34 OCS5_N 35 PRTPWR7 36 OCS7_N 37 OCS6_N 38 PRTPWR6 39 NON_REM1 40 CFG_SEL0 41 CFG_SEL1 42 RESET_N 43

VBUS_DET 44 NON_REM0 45

VDD33 46 PRT_SWP3 47 BOOST1 48 PRT_SWP2 49 BOOST0 50 PRT_SWP1 51

VDDA33_3 52 DN6_DM 53 DN6_DP 54 DN7_DM 55 DN7_DP 56

VDDA33_4 57 UP_DM 58 UP_DP 59

XTAL2 60

XTAL1 61 VDD18PLL 62 RBIAS 63 VDD33PLL 64 EP_VSS 65

U_ESD_UP

C3709087

IO1 1

IO2 2 GND 3 R_CC2

5.1kΩ

C_HUB_A1

100nF C_HUB_DD
100nF C_HUB_A2

100nF C_HUB_RESET 1uF C_HUB_CR_BULK

1uF C_HUB_A_BULK

1uF C_HUB_CR_HF 100nF C_XTAL1

18pF C_XTAL2

18pF

Y_HUB

C70590

X1 1 GND1 2

X2 3 GND2 4 R_CC1
5.1kΩ

J_DATA C165948

GND_A1 A1

VBUS_A4 A4 CC1 A5 DP_A6 A6 DM_A7 A7

SBU1 A8

VBUS_A9 A9 GND_A12 A12 GND_B1 B1 VBUS_B4 B4 CC2 B5 DP_B6 B6 DM_B7 B7

SBU2 B8

VBUS_B9 B9 GND_B12 B12

SHIELD SH

C_HUB_18

1uF C_HUB_18PLL 1uF C_HUB_A4

100nF C_HUB_A3

100nF C_HUB_PLL 100nF R_CFG0

10kΩ

R_CFG1

10kΩ

R_CFG2

10kΩ

R_SWAP4

100kΩ

R_NONREM1

10kΩ

R_SWAP3

100kΩ

R_SWAP5

100kΩ

R_SWAP7

100kΩ

R_SWAP6

100kΩ

R_SWAP2

100kΩ

R_DATA_OK1

10kΩ

C_DATA1

100nF U_DATA1 C11355

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 Q_DATA1 C85047

G

1
S

2 D

3

C_PWR2_IN

100nF C_DATA2

100nF U_PWR2 C3662799

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 U_DATA2 C11355

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10

J_PORT1 USB1130-15-A

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4

SHIELD 5 R_PWR_EN1

10kΩ

J_PORT2 USB1130-15-A

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4

SHIELD 5

C_PORT1_HF 100nF U_ESD1 C3709087

IO1 1

IO2 2 GND 3

Q_DATA2 C85047

G

1
S

2 D

3 C_PORT1_BULK

22uF C_PWR1_IN

100nF R_ILIM1
1.4kΩ

U_PWR1 C3662799

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10

C_PORT2_BULK

22uF C_PORT2_HF 100nF U_ESD2 C3709087

IO1 1

IO2 2 GND 3 R_DATA_OK2

10kΩ

C_PWR3_IN

100nF R_PWR_EN3

10kΩ

R_ILIM3

1.4kΩ

U_PWR3 C3662799

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10 Q_DATA4 C85047

G

1
S

2 D

3 C_PWR4_IN

100nF R_ILIM4
1.4kΩ

R_DATA_OK4

10kΩ

C_DATA4

100nF U_DATA4 C11355

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10R_PWR_EN4

10kΩ

U_PWR4 C3662799

EN_UVLO 1 OVLO 2
AUXOFF 3
FAULT_N 4
IN 5 OUT 6 DVDT 7 GND 8

ILM 9

ITIMER 10

Q_DATA3 C85047

G

1
S

2 D

3 R_PWR_EN2

10kΩ

R_ILIM2

1.4kΩ

J_PORT4 USB1130-15-A

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4

SHIELD 5 C_PORT4_HF 100nF U_ESD4 C3709087

IO1 1

IO2 2 GND 3 C_PORT4_BULK

22uF

R_DATA_OK3

10kΩ

C_DATA3

100nF C_PORT3_BULK

22uF

J_PORT3 USB1130-15-A

VBUS 1 D_MINUS 2 D_PLUS 3 GND 4
SHIELD 5 U_DATA3 C11355

VCC 1

SEL 2 D_PLUS 3 D_MINUS 4 GND 5 HSD1_MINUS 6 HSD1_PLUS 7 HSD2_MINUS 8 HSD2_PLUS 9 OE 10 U_ESD3 C3709087

IO1 1

IO2 2 GND 3 C_PORT3_HF 100nF

C_PWR_CTRL_OUT 1uF U_AND_PWR

C6053

1A 1
1B 2
1Y 3
2A 4
2B 5

2Y 6 GND 7

3Y 8

3A 9

3B 10

4Y 11

4A 12

4B 13

VCC 14 R_DATA_CMD4

10kΩ

R_PWR_CMD4

10kΩ

R_PWR_CMD3

10kΩ

R_PWR_CMD2

10kΩ

R_PWR_CMD1

10kΩ

R_DATA_CMD3

10kΩ

R_DATA_CMD2

10kΩ

R_DATA_CMD1

10kΩ

C_PWR_CTRL_IN

100nF R_ILIM_CTRL 187kΩ
C_PWR_CTRL_OUT_HF 100nF C_CTRL_VDD

100nF C_EXP_VDD

100nF C_CTRL_VUSB

330nF U_AND_DATA C6053

1A 1
1B 2
1Y 3
2A 4
2B 5

2Y 6 GND 7

3Y 8

3A 9

3B 10

4Y 11

4A 12

4B 13

VCC 14

C_AND_DATA 100nF C_AND_PWR

100nF

U_PWR_CTRL C130056

GND 1
IN1 2
IN2 3
EN 4

ILIM 5 OUT1 6 OUT2 7

FAULT_N 8

EP 9 R_GANG

10kΩ

U_EXP

C558584

GPB0 1 GPB1 2 GPB2 3 GPB3 4 GPB4 5 GPB5 6 GPB6 7 GPB7 8

VDD 9

VSS 10 NC1 11
SCL 12

SDA 13 NC2 14 A0 15

A1 16

A2 17 RESET_N 18

INTB 19

INTA 20 GPA0 21 GPA1 22 GPA2 23 GPA3 24 GPA4 25 GPA5 26 GPA6 27 GPA7 28 U_CTRL C130462

VDD 1 GP0 2 GP1 3 RST 4 URX 5 UTX 6 GP2 7 GP3 8

SDA 9

SCL 10

VUSB 11 D_MINUS 12 D_PLUS 13

VSS 14

R_BOOST0

10kΩ

R_BOOST1

10kΩ